

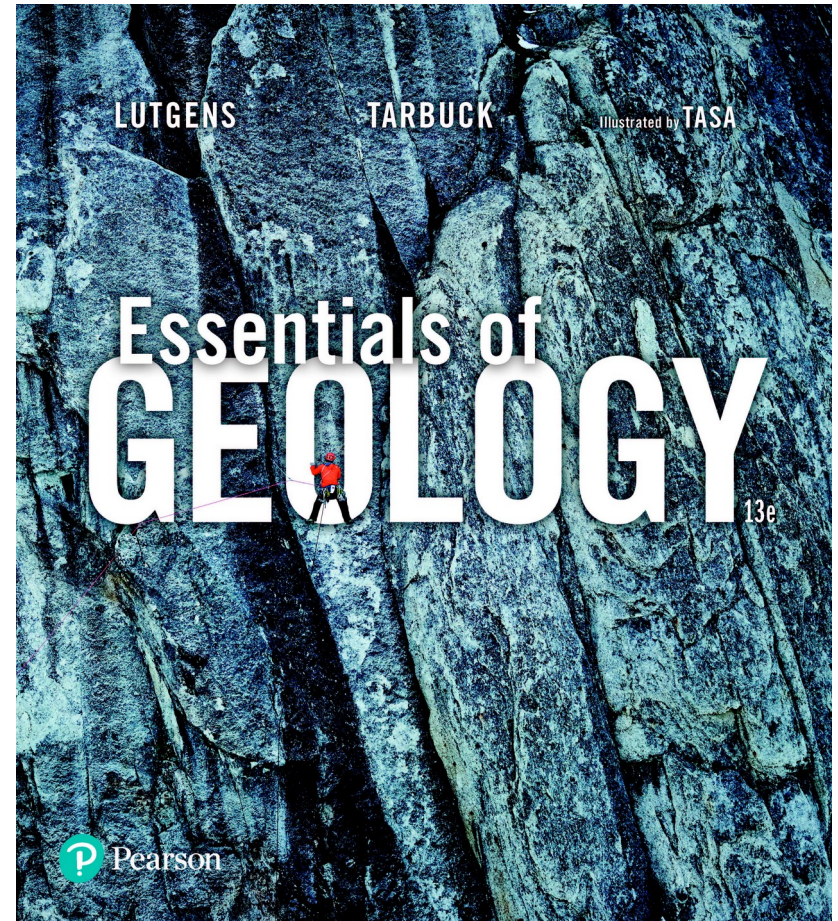
Lecture 2 Outline

Plate Tectonics: A Scientific Revolution Unfolds

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Essentials of Geology

Thirteenth Edition



Learning Objectives

- **LO 2.1** Summarize the view that most geologists held prior to the 1960s regarding the geographic positions of the ocean basins and continents.
- **LO 2.2** List and explain the evidence Wegener presented to support his continental drift hypothesis.
- **LO 2.3** List the major differences between Earth's lithosphere and asthenosphere and explain the importance of each in the plate tectonics theory.
- **LO 2.4** Sketch and describe the movement along a divergent plate boundary that results in the formation of new oceanic lithosphere.
- **LO 2.5** Compare and contrast the three types of convergent plate boundaries and name a location where each type can be found.

Learning Objectives (cont.)

- **LO 2.6** Describe the relative motion along a transform fault boundary and locate several examples of transform faults on a plate boundary map.
- **LO 2.7** Explain why plates such as the African and Antarctic plates are increasing in size, while the Pacific plate is decreasing in size.
- **LO 2.8** List and explain the evidence used to support the plate tectonics theory.
- **LO 2.9** Describe two methods researchers use to measure relative plate motions.
- **LO 2.10** Describe plate–mantle convection and explain two of the primary driving forces of plate motion.

2.1 From Continental Drift to Plate Tectonics

- Geologists thought the ocean basins and continents had fixed positions and were very old.
- A scientific revolution unfolded.

2.2 Continental Drift: An Idea Before Its Time

- Alfred Wegener
 - First proposed hypothesis, 1915
 - Published *The Origin of Continents and Oceans*
- Continental drift hypothesis
 - Supercontinent called Pangaea began breaking apart about 200 million years ago.
 - Continents “drifted” to present positions.
 - Continents “broke” through the ocean crust.

Pangaea Approximately 200 Million Years Ago

Tutorial

Modern reconstruction of Pangaea



Wegener's Pangaea, redrawn from his book published in 1915.



2.2 Continental Drift: An Idea Before Its Time

- Wegener's continental drift hypothesis
 - Evidence used by Wegener
 - Fit of South America and Africa
 - Fossils match across the seas
 - Rock types and structures match
 - Ancient climates
 - Main objection to Wegener's proposal was its inability to provide a mechanism.

Wegener's Matching of Mountain Ranges on Different Continents

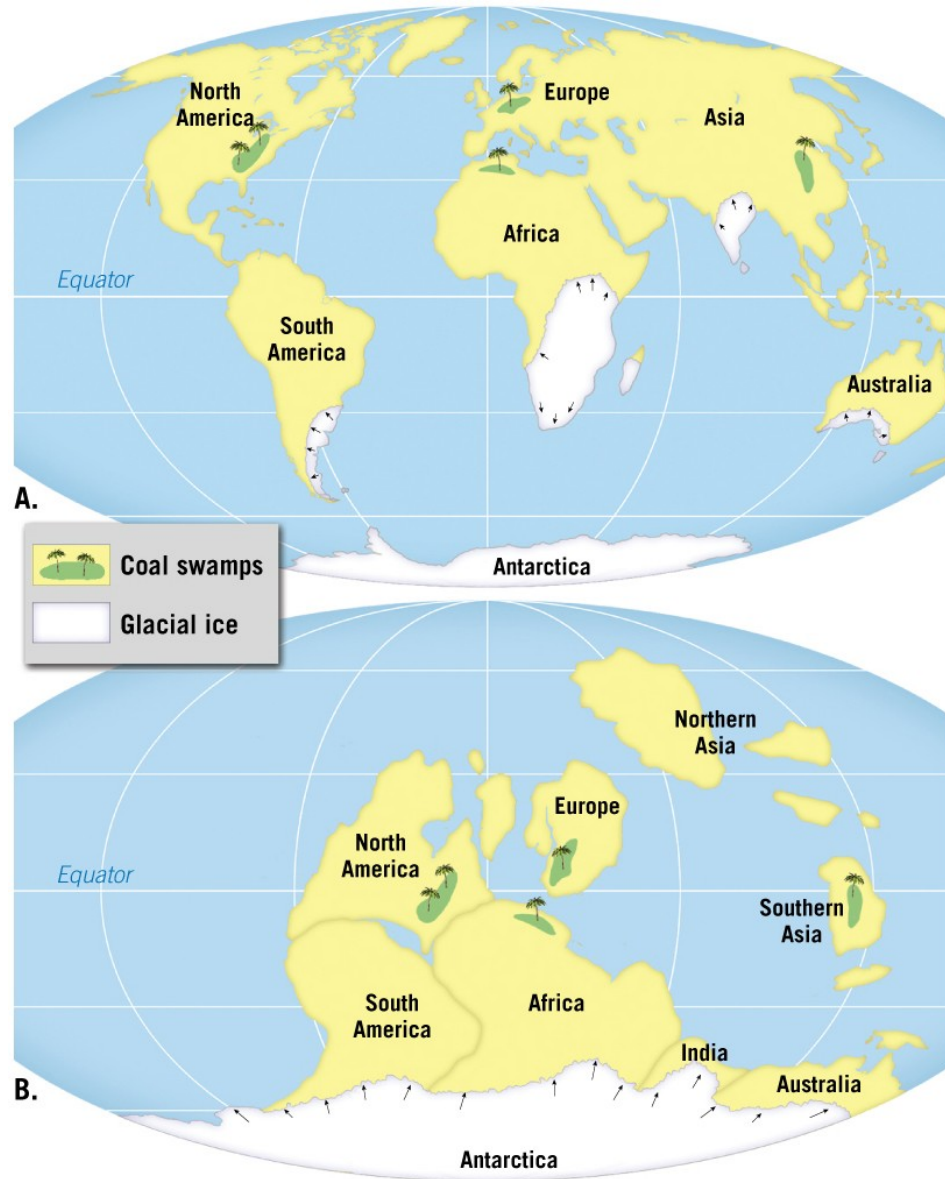


A.



B.

Paleoclimatic Evidence for Continental Drift



2.2 Continental Drift: An Idea Before Its Time

- Objections to the continental drift hypothesis
 - Wegener incorrectly suggested that the gravitational forces of the Moon and Sun were capable of moving the continents.
 - Wegener also incorrectly suggested that continents broke through the ocean crust.
 - There was strong opposition to this hypothesis from all areas of the scientific community.

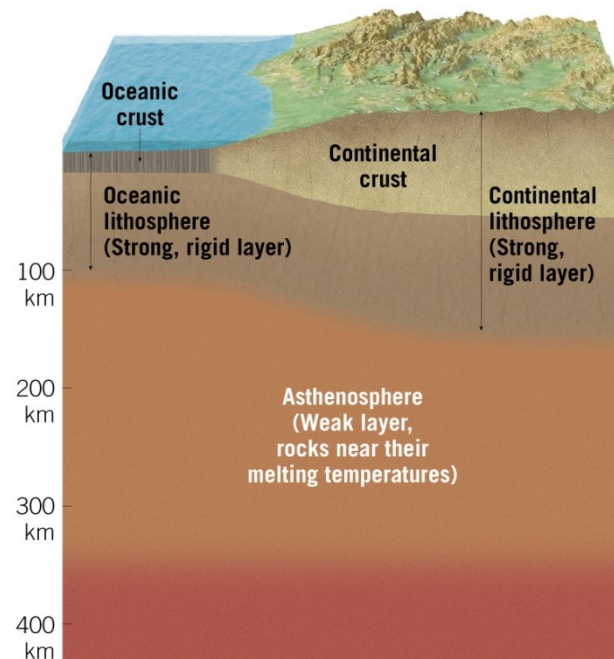
2.3 The Theory of Plate Tectonics

- More encompassing than continental drift
- Associated with Earth's rigid outer shell
 - Called the lithosphere
 - Consists of several plates
 - Plates are moving slowly
 - Largest plate is the Pacific plate
 - Plates are mostly beneath the ocean

2.3 The Theory of Plate Tectonics (cont.)

Tutorial

- Asthenosphere
 - Exists beneath the lithosphere
 - Hotter and weaker than lithosphere
 - Allows for motion of lithosphere



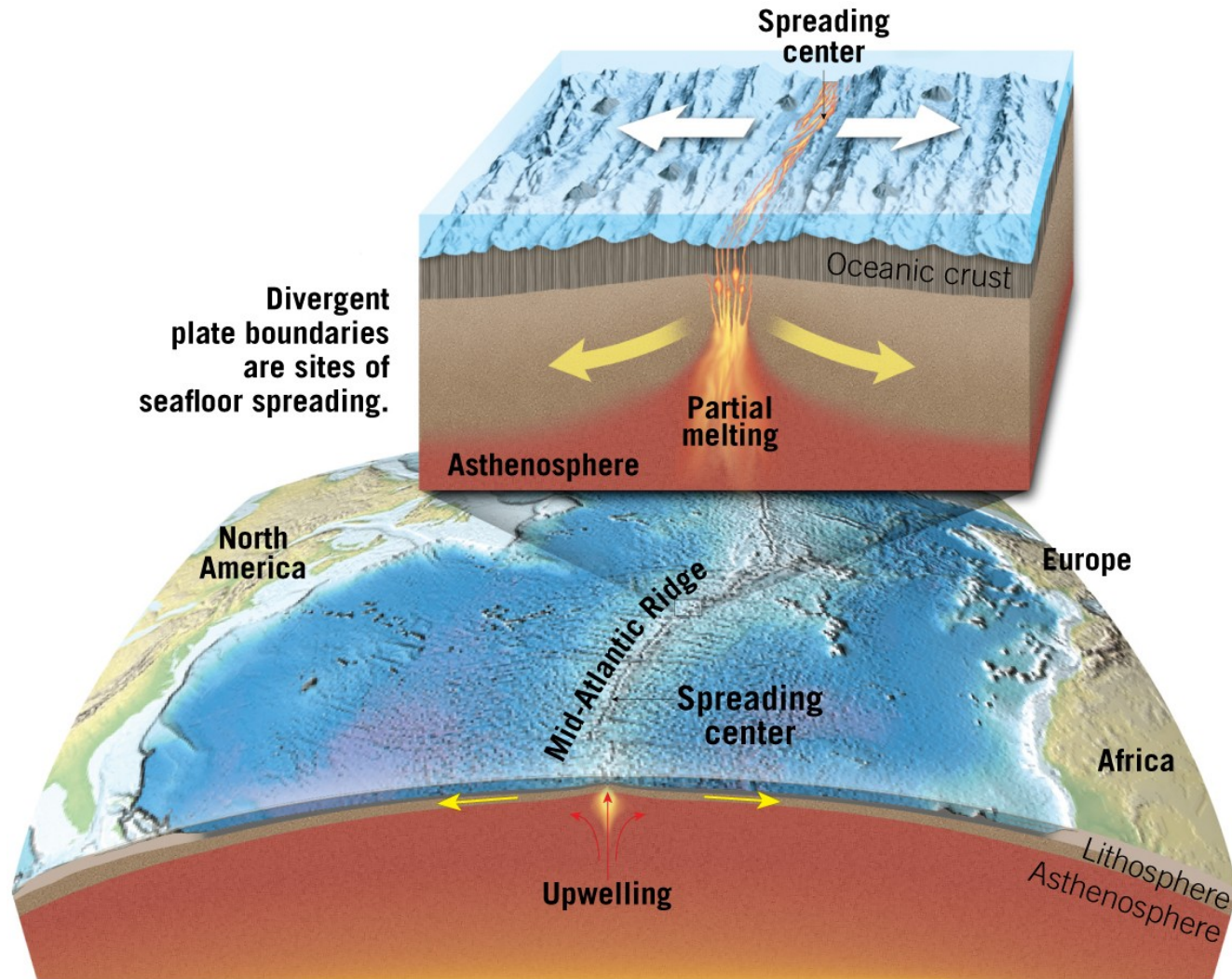
2.3 The Theory of Plate Tectonics (cont.)

- Plate boundaries
 - All major interactions among plates occur along their boundaries.
 - Types of plate boundaries
 - Divergent plate boundaries (constructive margins)
 - Continental rifts form at spreading centers within a continent.
 - Convergent plate boundaries (destructive margins)
 - Plates collide, an ocean trench forms, and lithosphere is subducted into the mantle.
 - Transform plate boundaries
 - Plates slide past one another and no lithosphere is created or destroyed.

2.4 Divergent Plate Boundaries and Seafloor Spreading

- Divergent plate boundaries (constructive margins)
 - Two plates move apart.
 - Mantle material upwells to create new seafloor.
 - Ocean ridges and seafloor spreading
 - Oceanic ridges develop along well-developed boundaries.
 - Along ridges, seafloor spreading creates new seafloor.
 - Can also occur in continental crust

2.4 Divergent Plate Boundaries and Seafloor Spreading (cont.)



Rift Valley in Iceland

Mobile Field Trip

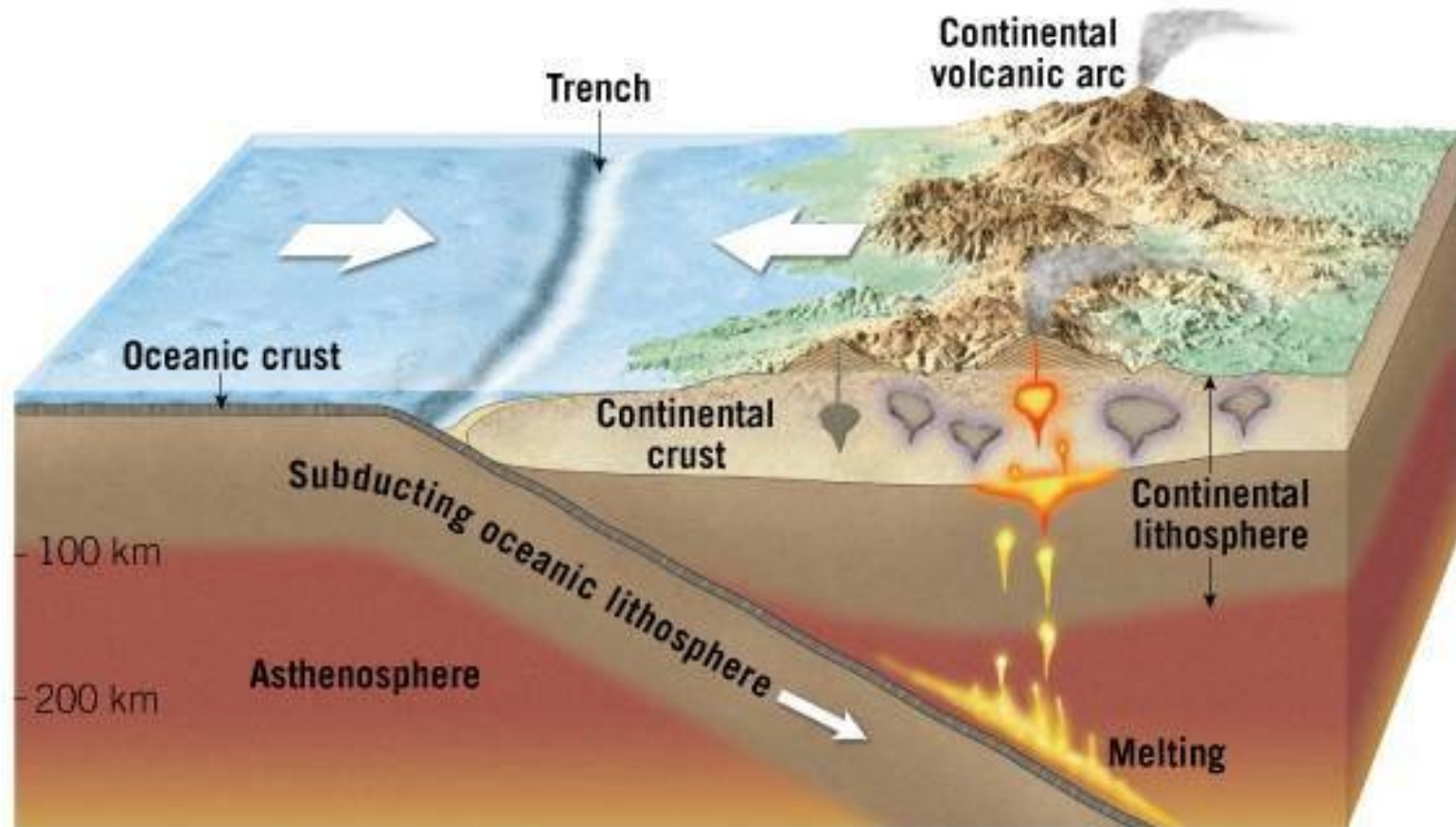


2.5 Convergent Plate Boundaries and Subduction

- Convergent plate boundaries (destructive margins)
 - Oceanic–continental convergence
 - Denser oceanic slab sinks into the asthenosphere.
 - Pockets of magma develop and rise.
 - Continental volcanic arcs form.
 - Examples include the Andes, Cascades, and the Sierra Nevadan system.

An Oceanic–Continental Convergent Plate Boundary

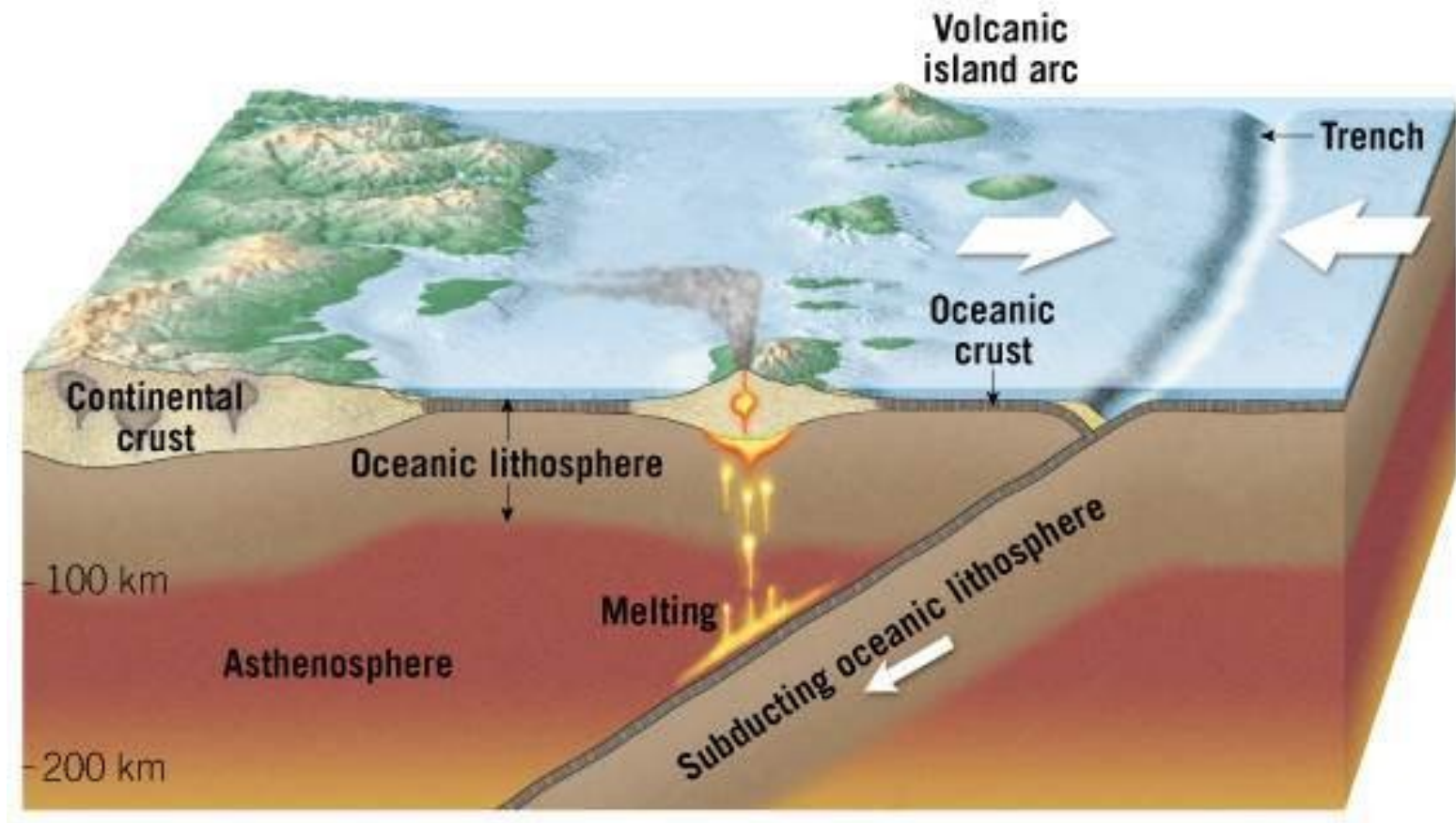
[Tutorial](#)



2.5 Convergent Plate Boundaries and Subduction

- Convergent plate boundaries (destructive margins)
 - Oceanic–oceanic convergence
 - Two oceanic slabs converge and one descends beneath the other.
 - Often forms volcanoes on the ocean floor.
 - Volcanic island arcs form as volcanoes emerge from the sea.
 - Examples include the Aleutian, Mariana, and Tonga islands.

An Oceanic–Oceanic Convergent Plate Boundary

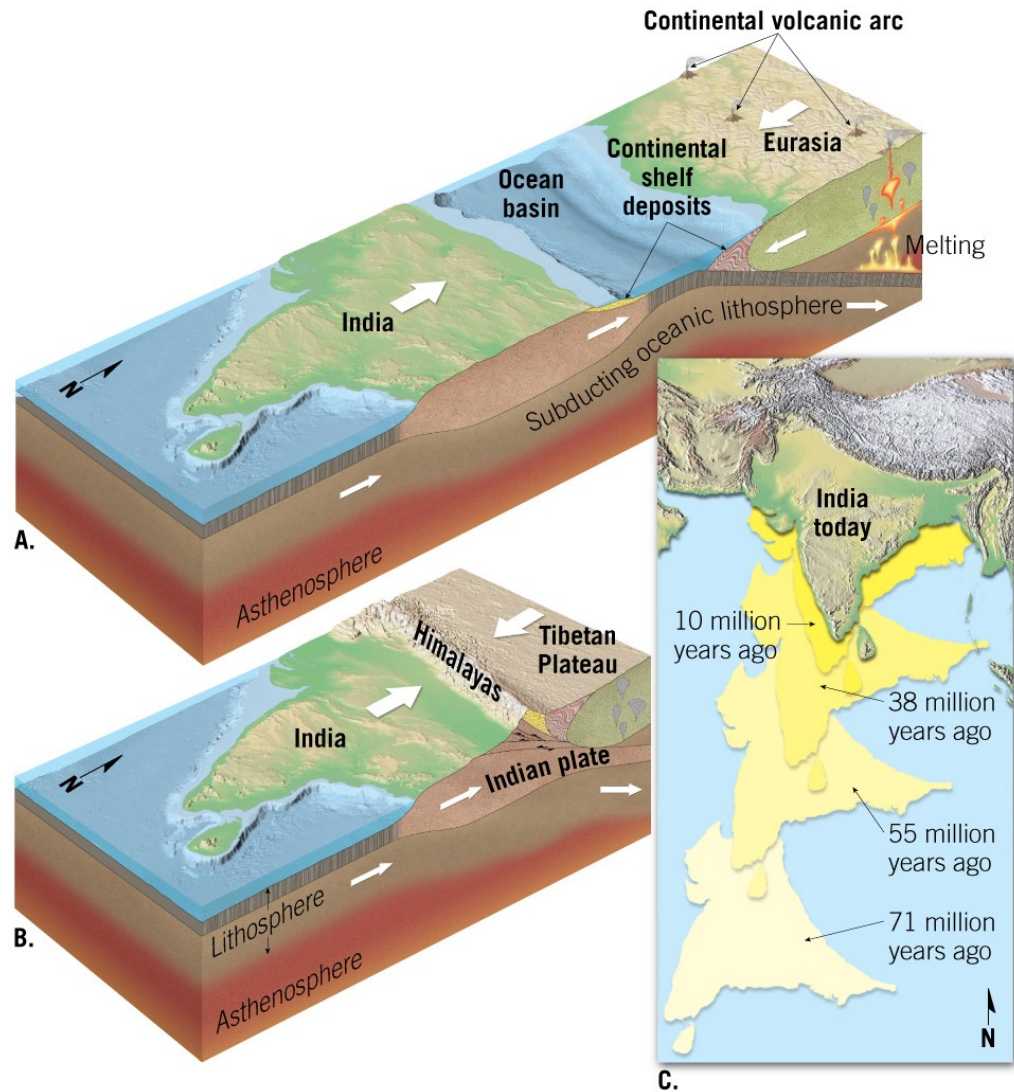


2.5 Convergent Plate Boundaries and Subduction

- Convergent plate boundaries (destructive margins)
 - Continental–continental convergence
 - When subducting plates contain continental material, two continents collide.
 - Can produce new mountain ranges such as the Himalayas.

The Collision of India and Asia Produced the Himalayas (Before)

[Animation](#)



2.6 Transform Plate Boundaries

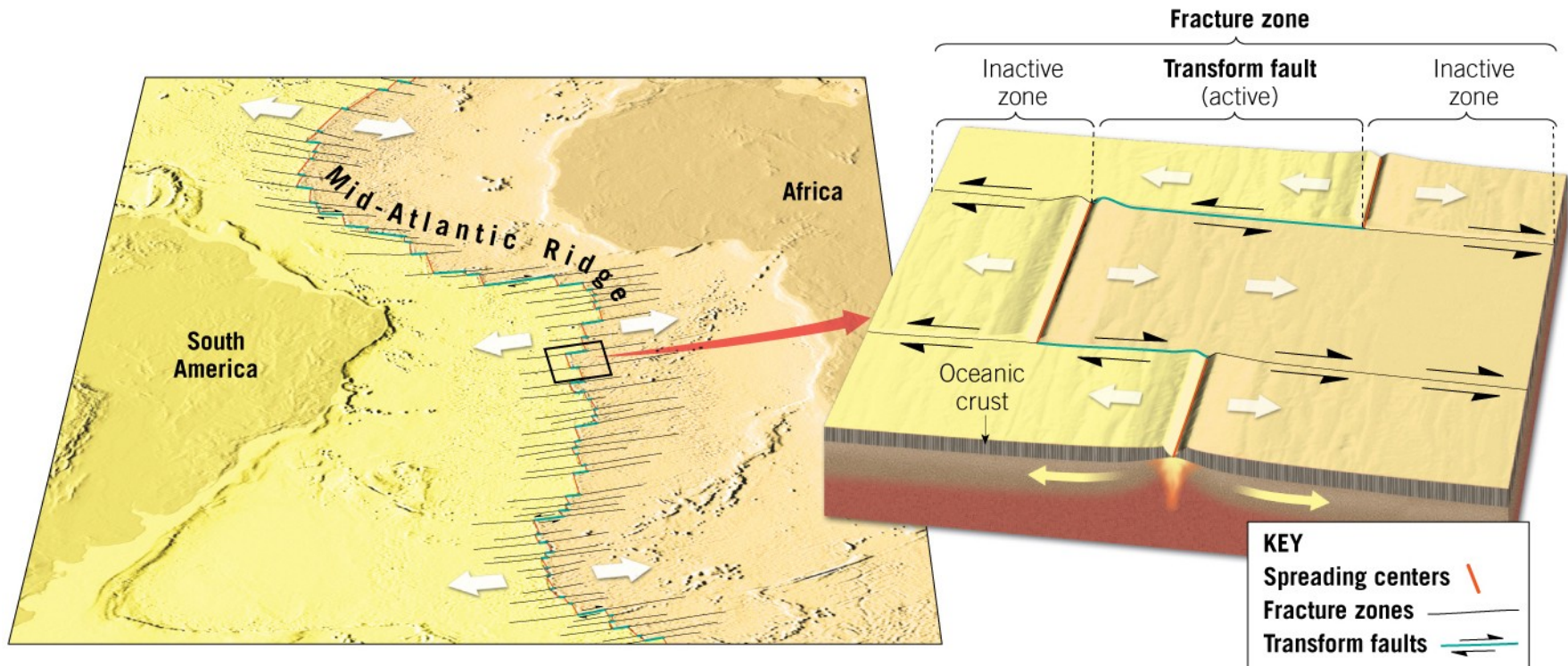
- Transform Plate Boundaries
 - Plates slide past one another and no lithosphere is created or destroyed.
 - Most join two segments of an oceanic ridge system along breaks in the oceanic crust known as fracture zones.
 - Transform faults can also move ridge crests toward subduction zones.
 - A few transform faults cut through continental crust.
 - Examples include:
 - The San Andreas Fault
 - The Alpine Fault of New Zealand

2.6 Transform Plate Boundaries (cont.)

Tutorial

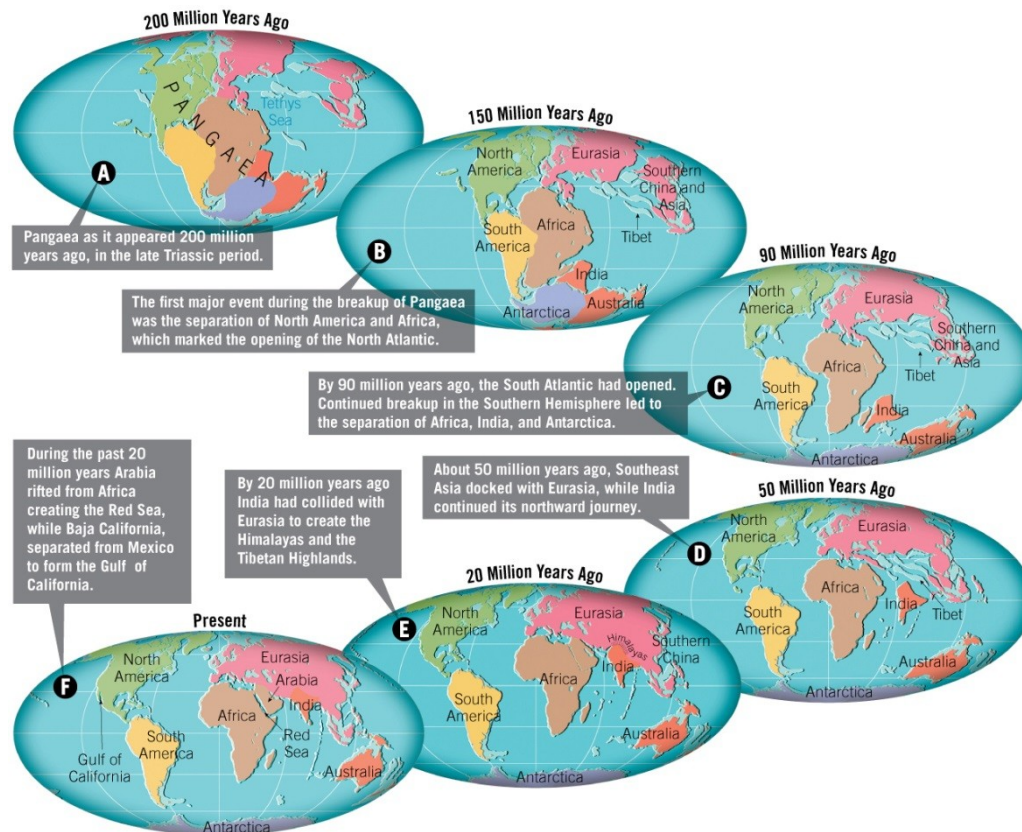
A. The Mid-Atlantic Ridge, with its zigzag pattern, roughly reflects the shape of the rifting zone that resulted in the breakup of Pangaea.

B. Fracture zones are long, narrow scar-like features in the seafloor that are roughly perpendicular to the offset ridge segments. They include both the active transform fault and its “fossilized” trace.



2.7 How Do Plates and Plate Boundaries Change?

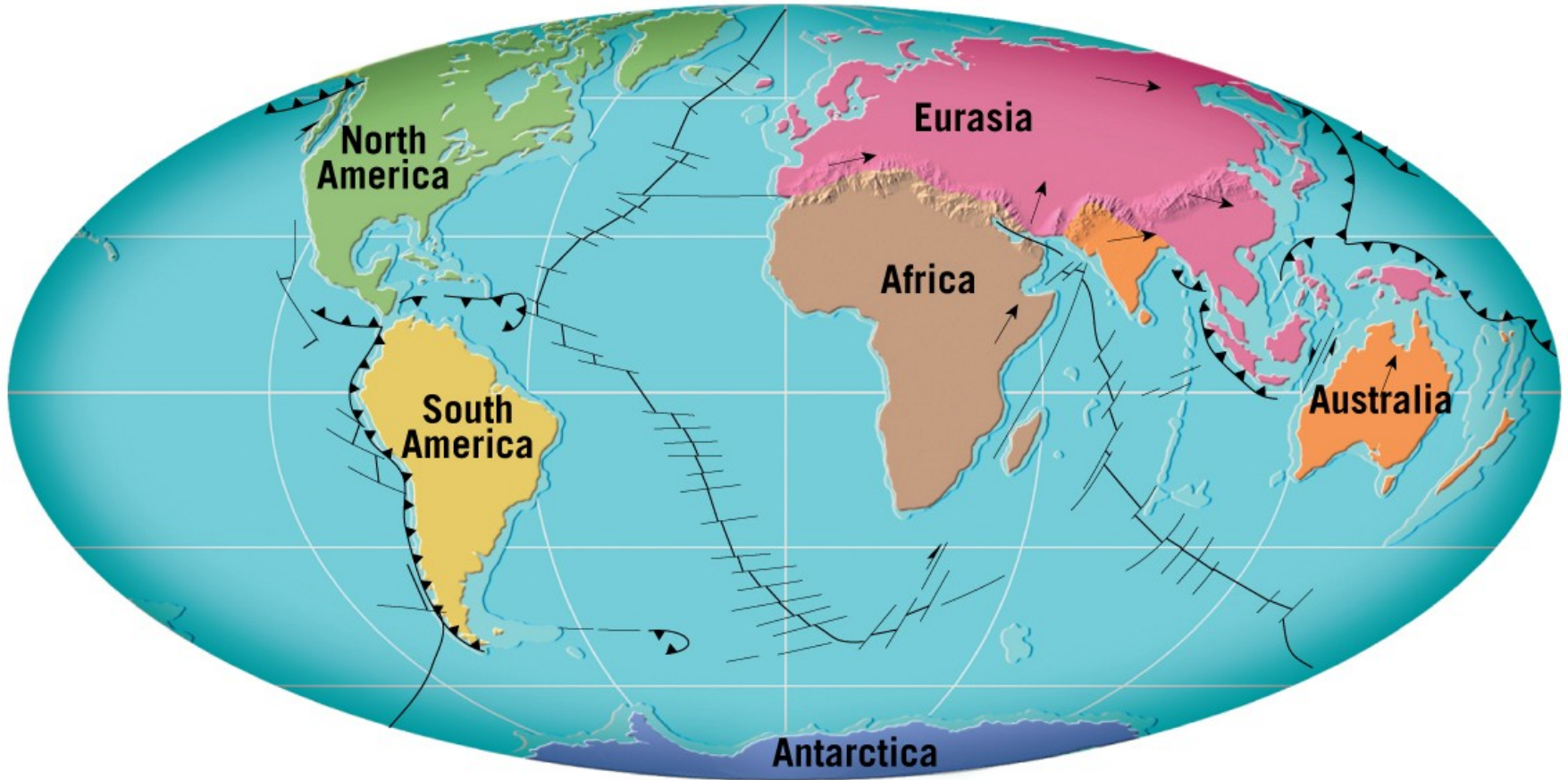
- Earth's total surface area does not change
 - Pangaea existed 200 million years ago



2.7 How Do Plates and Plate Boundaries Change? (cont.)

- Present-day motions have been extrapolated into the future some 50 million years
 - Areas west of the San Andreas Fault slide northward past the North American plate.
 - Africa collides with Eurasia, closing the Mediterranean and initiating mountain-building.
 - Australia and new Guinea are on a collision course with Asia.

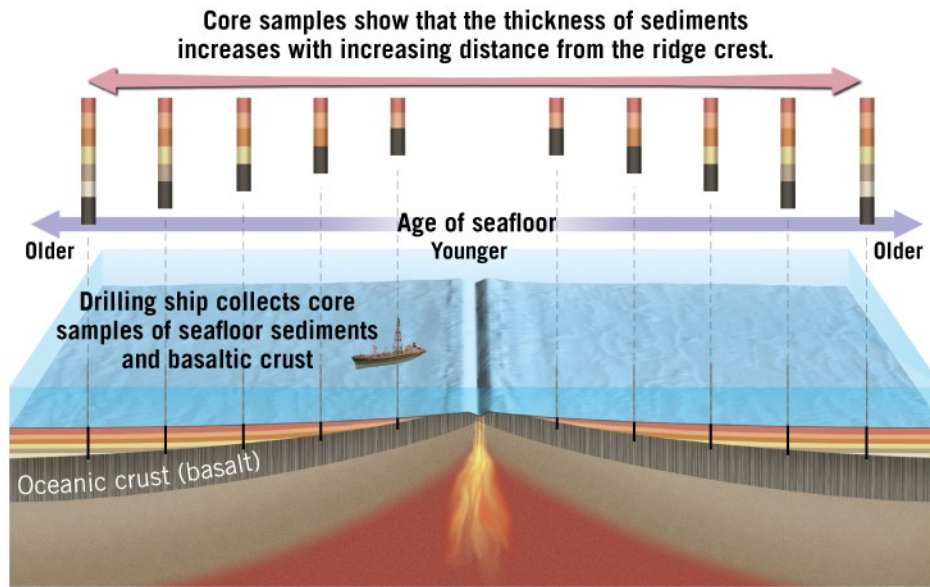
A Possible View of the World 50 Million Years from Now



2.8 Testing the Plate Tectonics Model

- Evidence from ocean drilling
 - Some of the most convincing evidence confirming seafloor spreading has come from drilling directly into ocean-floor sediment.
 - Age of deepest sediments
 - Thickness of ocean-floor sediments verifies seafloor spreading.

Evidence from Ocean Drilling



A.

Chikyu is a state-of-the-art drilling ship designed to drill up to 7000 meters (more than 4 miles) below the seafloor.

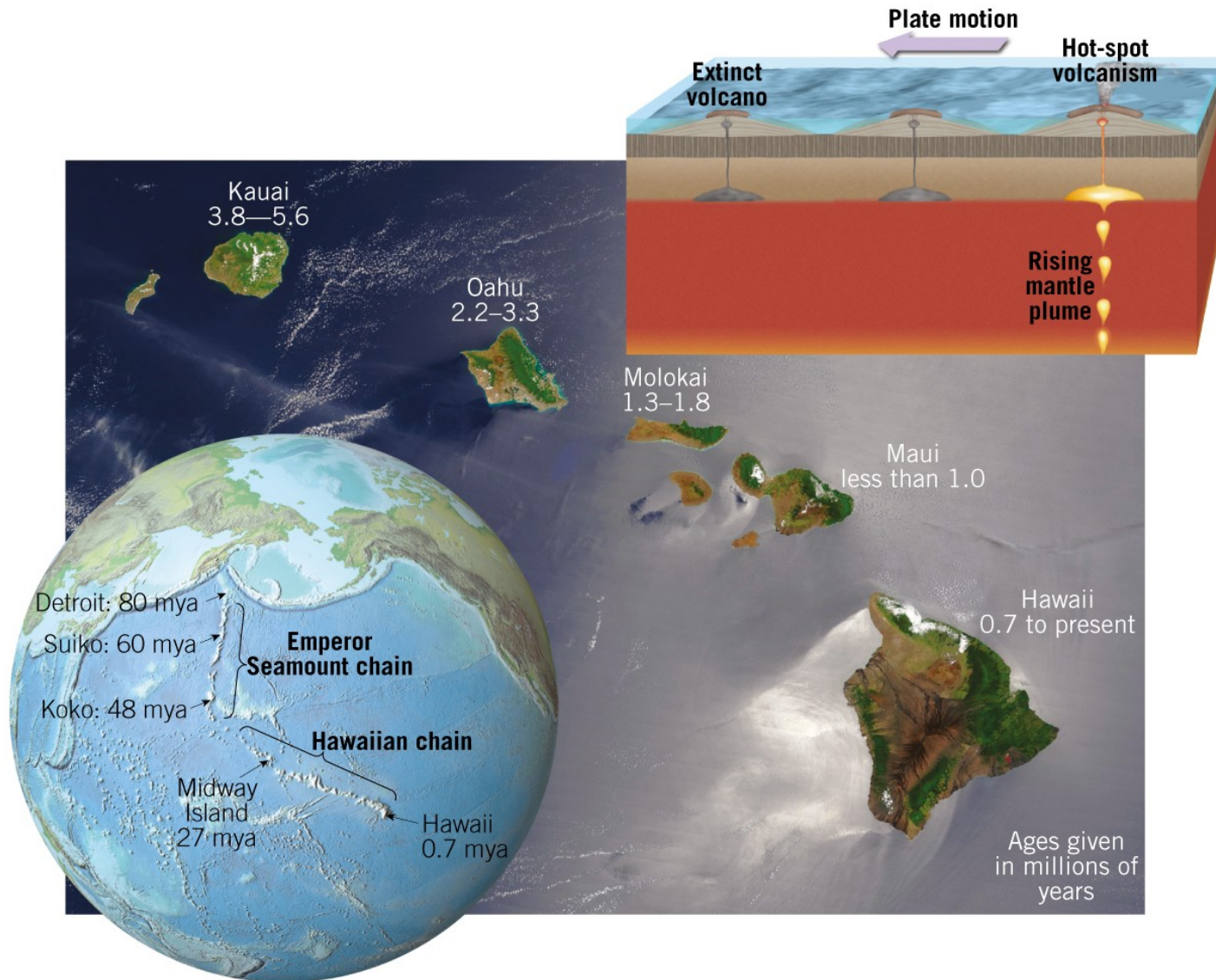


B.

2.8 Testing the Plate Tectonics Model

- Hot spots and mantle plumes
 - Caused by rising plumes of mantle material.
 - Volcanoes can form over them (Hawaiian island chain).
 - Mantle plumes
 - Long-lived structures
 - Some originate at great depth, perhaps at the mantle-core boundary.

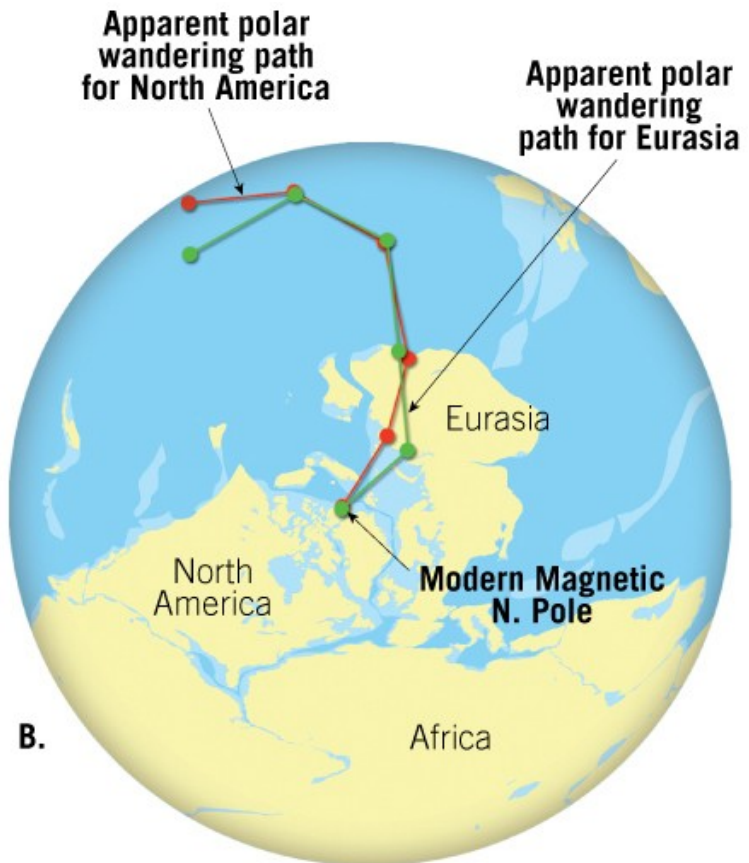
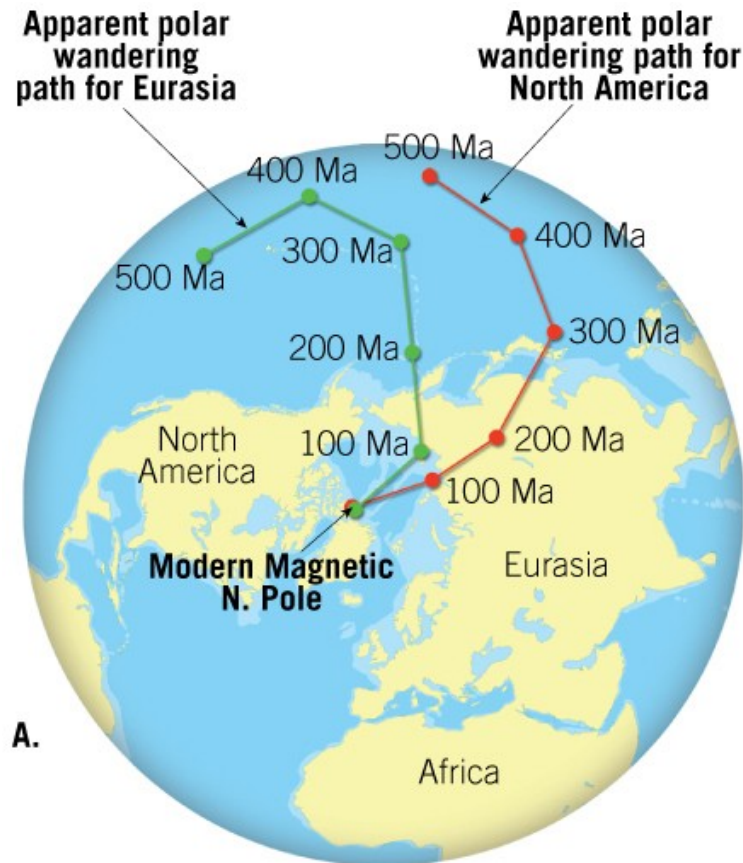
Hot Spots and Hot-Spot Tracks



2.8 Testing the Plate Tectonics Model

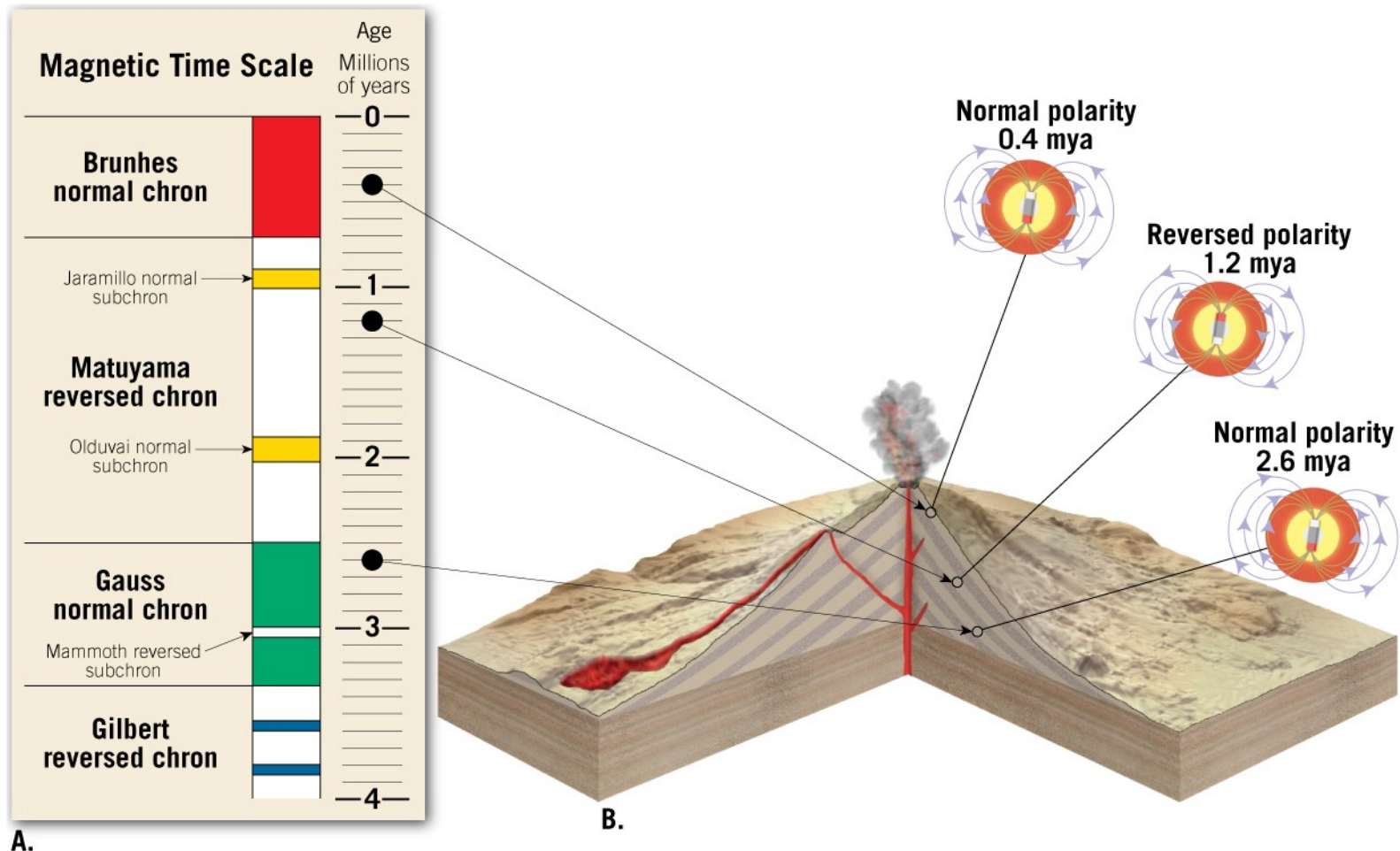
- Evidence for the plate tectonics model
 - Paleomagnetism
 - Probably the most persuasive evidence
 - Ancient magnetism preserved in rocks
 - Paleomagnetic records show
 - Polar wandering (evidence that continents moved)
 - Earth's magnetic field reversals
 - Recorded in rocks as they form at oceanic ridges

Apparent Polar-Wandering Paths for Eurasia and North America



Paleomagnetic Reversals Recorded by Basalt Flows at Mid-Ocean Ridges

Tutorial

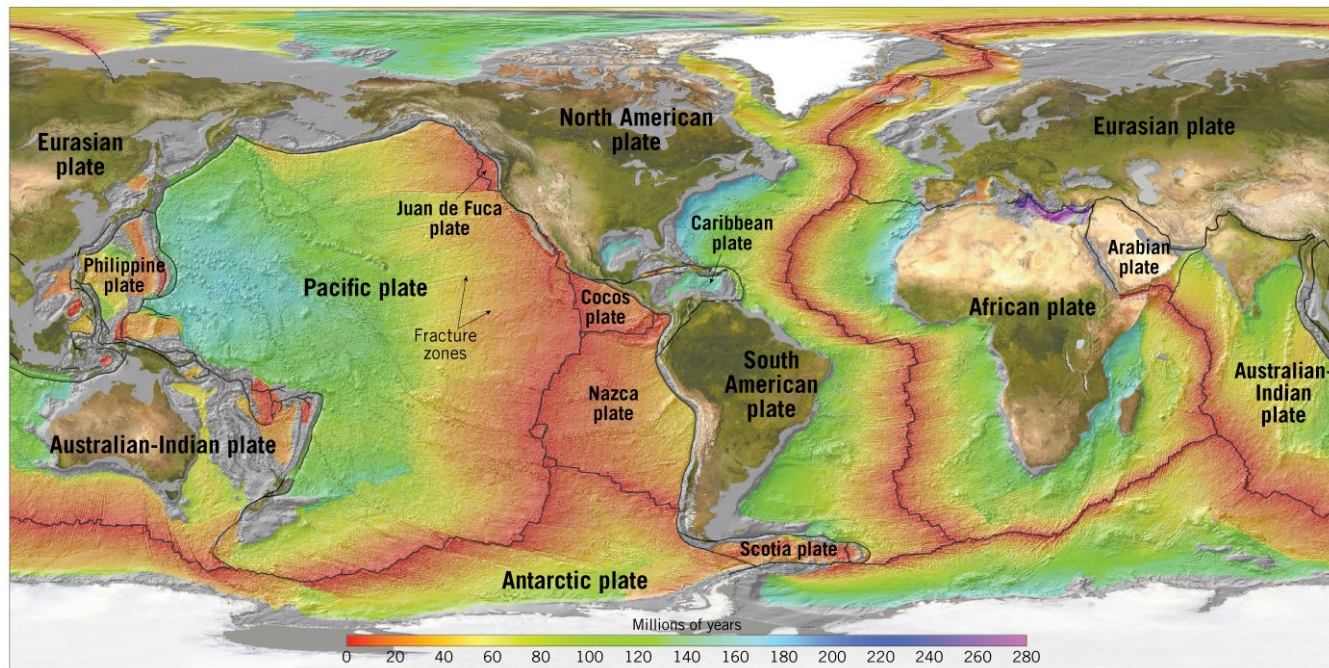


2.9 How Is Plate Motion Measured?

- By using hot-spot “tracks,” like those of the Hawaiian Island–Emperor Seamount chain, and the age of the seafloor
- Using space-age technology to directly measure the relative motion of plates
 - Very Long Baseline Interferometry (VLBI)
 - Global Positioning System (GPS)

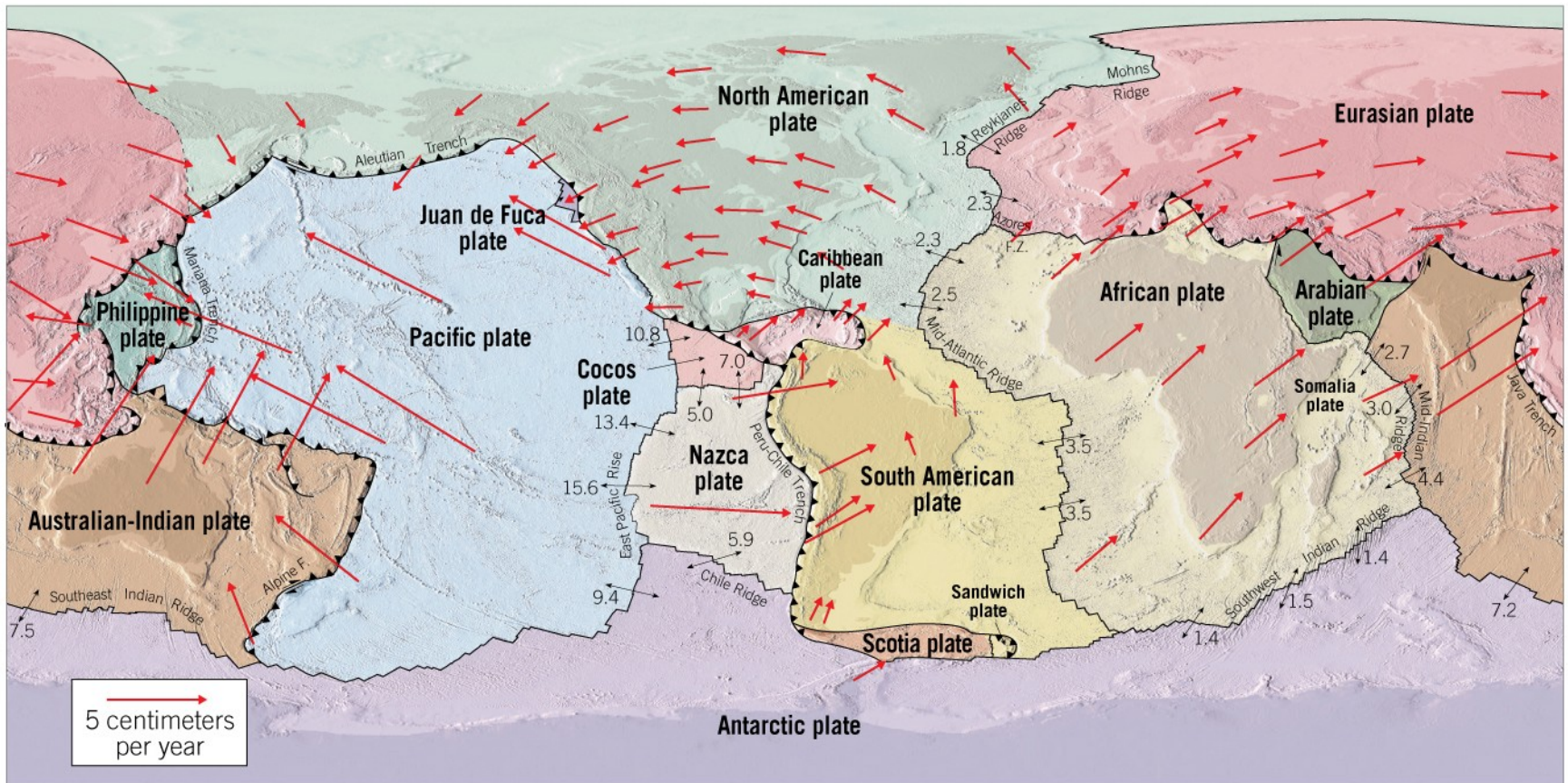
2.9 How Is Plate Motion Measured? (cont.)

- Geologic Evidence for Plate Motion
 - By knowing the age of the seafloor and the distance from the spreading center, an average rate of plate motion can be calculated



Directions and Rates of Plate Motions

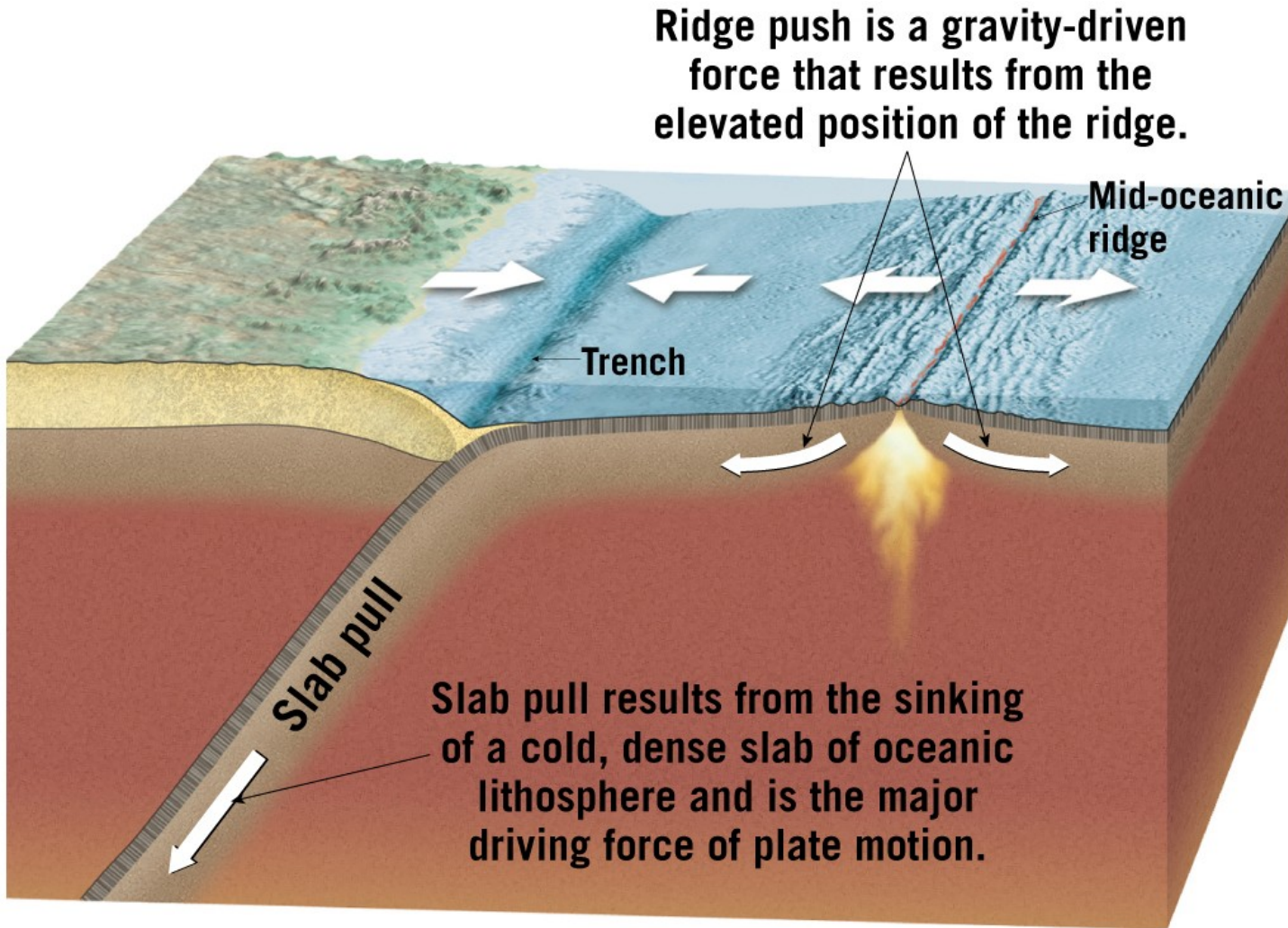
Directions and rates of plate motions measured in centimeters per year



2.10 What Drives Plate Motion?

- Driving mechanism of plate tectonics
 - No one model explains all facets of plate tectonics.
 - Earth's heat is the driving force through convection.
 - Several models have been proposed.
 - Slab-pull and ridge-push model
 - Descending oceanic crust pulls the plate.
 - Elevated ridge system pushes the plate.

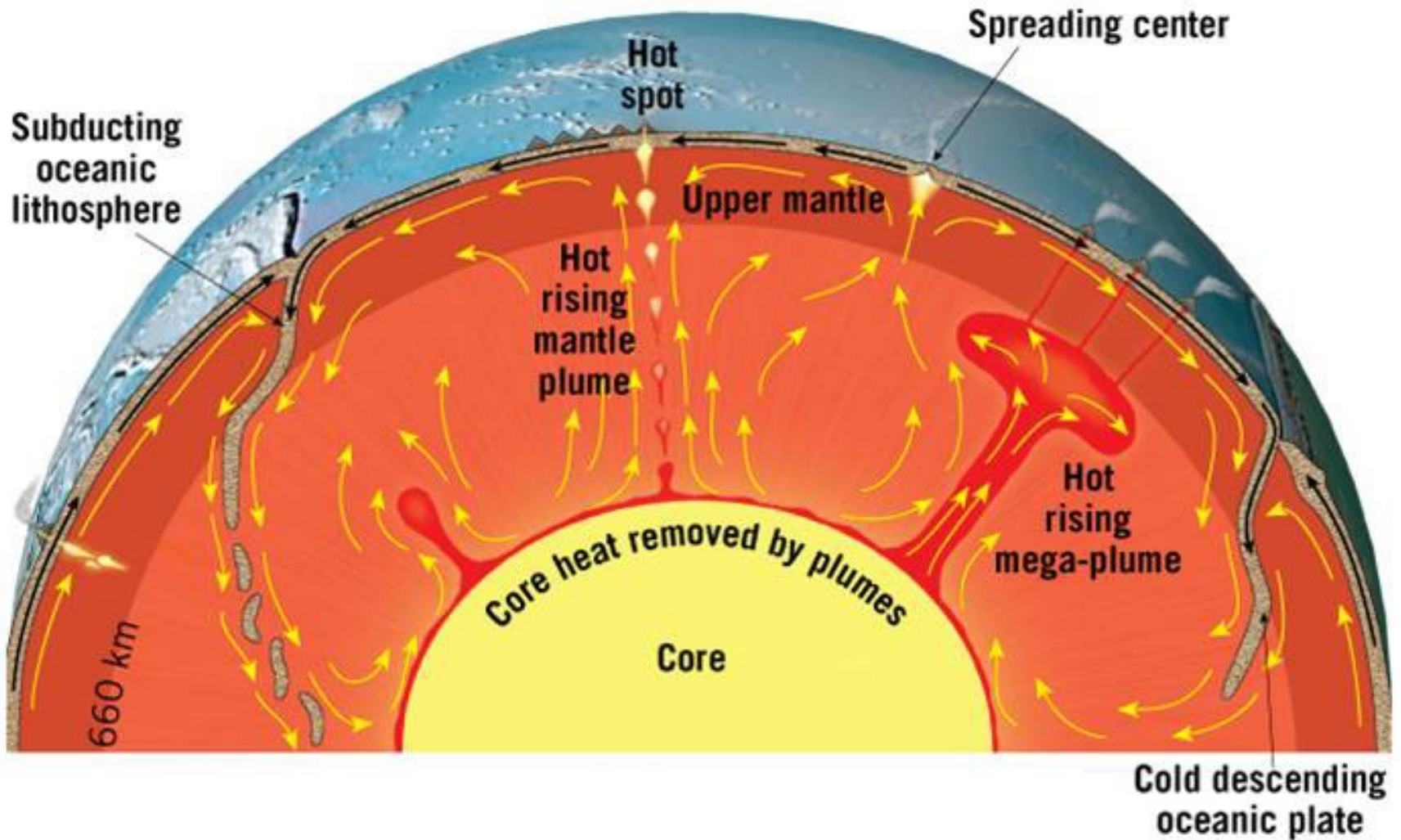
Mechanisms Contributing to Plate Motion



2.10 What Drives Plate Motion?

- Several models have been proposed
 - Plate–mantle convection
 - Mantle plumes extend from mantle-core boundary and cause convection within the mantle
 - Models
 - Whole-mantle convection
 - “Layer Cake” Model

Whole-Mantle Convection



“Layer Cake Model”

